

# Curriculum Vitae

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## PERSONAL INFORMATION

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**Name** **Gaurav Sablok**  
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## ACADEMIC RESEARCH PROFILE AND STATEMENT OF EXPERTISE

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My research and development activities span the interdisciplinary fields of bioinformatics, computational genomics, machine learning, deep learning, and scientific software development, with applications across a broad range of biological species and sequencing technologies. My primary research focus encompasses both Illumina short-read and long-read sequencing technologies, including the analysis of genomic, transcriptomic, metagenomic, and organellar datasets. In parallel, I develop computational methods, bioinformatics software, machine-learning and deep-learning models, and web-based scientific applications to address complex biological and genomic questions.

My principal programming languages are Python and Rust. I primarily use Python for data analysis, machine learning, deep learning, and rapid development of computational methods. Rust constitutes an increasingly important component of my software-development strategy, particularly for high-performance bioinformatics, machine-learning infrastructure, standalone scientific applications, and full-stack web development. My background as a C++ developer, spanning approximately a decade, has strongly influenced my preference for Rust as a modern systems-programming language that combines performance, memory safety, and expressive software engineering.

I aim to maintain a unified and productive development environment by using Rust and Python according to their respective strengths: Python enables rapid prototyping, data analysis, and the development and evaluation of machine- and deep-learning models, whereas Rust enables the development of efficient, reliable, and scalable production software. This combination allows me to move from exploratory, scripting-based analyses to standalone, high-throughput computational systems without extensive dependence on multiple programming languages, and I have applied this philosophy across bioinformatics, machine learning, deep learning, transformer-based approaches, web development, and scientific application development.

## WORK EXPERIENCE

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### 2025 – 2025

**Area Expert (Bioinformatics)** | *Institute of Bioorganic Chemistry, Polish Academy of Sciences, Poznań, Poland*

Worked primarily on human genomics, developing computational approaches and software for the analysis and interpretation of human genomic data. Extended experience in genome-scale computational analysis and strengthened the ability to translate biological research questions into reproducible computational methods and software solutions.

### 2024 – 2024

**Academic Staff Member** | *University of Potsdam, Germany*

Developed computational approaches related to pangenomics, machine learning and deep learning. Performed PacBio HiFi genome analyses for pangenomic applications and contributed to the development of a complete web-based scientific application using HTML, CSS and JavaScript.

### Jul 2017 – Dec 2021

**Postdoctoral Researcher** | *Finnish Museum of Natural History (LUOMUS) and University of Helsinki, Finland*

Conducted research in genome bioinformatics, comparative genomics, genome assembly, genome annotation, chloroplast genomics, phylogenomics and high-performance computing. Contributed to the sequencing, assembly, annotation and comparative analysis of genomes from a diverse range of lower plants and other organisms, including *Coleochaete orbicularis*, *Blasia pusilla*, *Chaetosporidium orbicularis*, *Polytrichum commune*, *Mallomonas* and *Cryptomonas* species. Developed bioinformatics workflows suitable for high-performance computing and museum-based genomic research, including genomic analyses for PAFTOL species (Edinburgh, UK) and chloroplast genomes of the *Ambrosia* clade (Norway). From ~2019, research increasingly focused on fungal genomics using Illumina NextSeq data; assembled and annotated fungal genomes from more than 500 species, including ITS and other phylogenomic marker analyses, sequence alignment, phylogenetic reconstruction and comparative genomics.

### Aug 2016 – Jan 2017

**University Fellow** | *University of Connecticut, Storrs, USA*

Conducted computational analyses of the Douglas-fir genome, covering genome annotation, comparative genomics, phylogenomics, gene identification and the investigation of genes and evolutionary processes of biological significance.

**Sept 2014 – Sept 2016**

**Research Fellow** | *University of Technology Sydney (UTS), Australia*

Developed computational methods for understanding seagrass genomes and biological systems. Developed and applied computational approaches for cluster and high-performance computing environments, strengthening experience in large-scale scientific computation.

**Sept 2011 – Sept 2014**

**Postdoctoral Researcher (CoCoPro)** | *Fondazione Edmund Mach (FEM), IASMA, Italy*

Developed bioinformatics methods and software for transcriptional and post-transcriptional genomics across nuclear and organellar genomes. Analysed and completed multiple RNA-seq and organelle-sequencing projects across several plant and fungal species, including *Arundo donax*. Conducted extensive metagenomic analyses of fungal and bacterial communities (ITS-based fungal metagenomics and bacterial metagenomics). Developed expertise in organelle genomics, particularly chloroplast genomics, and contributed to the publication of the first chloroplast genomes from Cardamine species.

**Dec 2010 – Aug 2011**

**Visiting Postdoctoral Researcher** | *Huazhong Agricultural University, China*

## EDUCATION AND TRAINING

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**2022 – 2023**

**Career Addition Certification — Deep Learning & Data Science**

Continuing professional development in deep learning and data science methodologies.

**Academic Appointments (Chronological Summary)**

- 2025–2025: Bioinformatics – Area Expert, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Poland.
- 2024–2024: Academic Staff Member, University of Potsdam, Germany.
- 2022–2023: Career Addition — Deep Learning, Data Science Certification.
- Jul 2017 – Dec 2021: Postdoctoral Researcher, Finnish Museum of Natural History, Finland.
- Aug 2016 – Jan 2017: University Fellow, University of Connecticut, Storrs, USA.
- Sept 2014 – Sept 2016: Research Fellow, University of Technology Sydney, Australia.
- Sept 2011 – Sept 2014: CoCoPro, Fondazione Edmund Mach, IASMA, Italy.
- Dec 2010 – Aug 2011: Visiting Postdoctoral Researcher, Huazhong Agricultural University, China.

## FELLOWSHIPS AND TRAVEL GRANTS

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- 2014–2016: UTS Internal Start-up Grant (AUD 20K) — Project Role: Principal Investigator.
- 2015–2016: KARORA — Graphic-driven platform-independent proteomics workflow for spectral libraries. University of Technology Sydney (AUD 15K) — Project Role: Principal Investigator.
- 2012–2014: CoCoPro Fellowship — "Developing NGS tools and techniques for large-scale methods implementation in transcriptomics and genomics."
- 2011–2012: EU Postdoctoral Fellowship (BIOMASFOR) — "Enhancing biomass potential for European forest using NGS technologies."
- 2012: Travel grant, European Conference on Computational Biology (ECCB), Swiss Institute of Bioinformatics — for presenting FungPROTDB: Database of protein homo-repeats in fungal proteomes and secretomes.
- 2010–2011: Chinese Fellowship — "Developing transcriptomics tools for *Diospyros kaki*."

## PERSONAL SKILLS

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**Research and Technical Expertise**

- Genome assembly and genome annotation
- Comparative genomics and pangenomics
- Phylogenomics and evolutionary genomics
- Chloroplast and other organellar genomics
- RNA-seq and transcriptomics

- Metagenomics
- Illumina and long-read sequencing technologies
- PacBio HiFi genome analysis
- High-performance computing and scientific computing
- Bioinformatics workflow and software development
- Machine learning and deep learning
- Transformer-based computational approaches
- Scientific web and application development

## **Programming Languages**

Python (data analysis, machine learning, deep learning, rapid prototyping) · Rust (high-performance computing, ML infrastructure, full-stack web development) · C++ (approx. 10 years' systems-programming background)

## **SUITABILITY AND RESEARCH INTERESTS**

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This combination of experience makes me a strong and versatile candidate for computational genomics and bioinformatics research positions. My research trajectory extends from fundamental genome analysis and bioinformatics through computational method development, machine learning, deep learning, and scientific software engineering — contributing across multiple stages, from raw sequencing data and genome assembly through biological interpretation, computational modelling, and the development of production-ready software.

A distinctive aspect of my profile is the combination of bioinformatics expertise with modern software engineering and machine/deep learning. My experience with Python enables the rapid development and evaluation of computational and machine-learning models, while my expertise in Rust allows me to develop efficient, scalable, and maintainable software for high-throughput scientific applications, building on a strong foundation in systems programming from my earlier work with C++.

My research interests are directly aligned with computational genomics, bioinformatics, machine learning, and scientific software development. I value collaborative, productive, and interdisciplinary research environments and seek to translate complex scientific questions into robust computational solutions that generate reproducible and biologically meaningful results.

## **GRANT REVIEWER**

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- Ohio Agricultural Research and Development Center (OARDC) (2014), Ohio Plant Biotechnology Consortium, USA.
- Reviewer, WHITE proposal (2018), Chargée de Projets Scientifiques, Agence Nationale de la Recherche, France.

### Peer-Reviewed Publications (\$ Corresponding Author; \* Contributed Equally)

1. Sablok G. (2026). Coffee Genomics: Recent Approaches Toward Improved Cup Quality and Climate Resilience. Preprints. <https://doi.org/10.20944/preprints202609.1301.v1>
2. Sablok G. (2026). Pangenomic Graph Softwares as Passport for Shared Global Applicability. Preprints. <https://doi.org/10.20944/preprints202609.0994.v1>
3. Sablok G. (2026). eVaiutilities, evaifilter and evaiweb: Genomic Population-Scale Utilities for eVai Analysis. Preprints. <https://doi.org/10.20944/preprints202609.0847.v1>
4. Sablok G. (2026). phyevkit: Post-Alignment Toolkit for Phylogenomics. Preprints. <https://doi.org/10.20944/preprints202609.0745.v1>
5. Sablok, G. (2026). Bottleneck and Perspectives on Machine Learning and Artificial Intelligence in Plant Microbiome Research. Preprints. <https://doi.org/10.20944/preprints202609.1390.v1>
6. Pewinska-Kolodziejczak M, Kotowska-Zimmer A, Przybyl L, Wronka D, Karlik A, Smuszkiewicz M, Balcerek J, Suszynska-Zajczyk J, Piaszyk-Borychowska A, Urbanek-Trzeciak M, Sablok G, Kozłowska E, Podkowinski J, Fiszler A, Kozłowski P, Handschuh L, Olejniczak M (2026). CAG-targeting artificial miRNA with reduced off-target risk for efficient lowering of pathogenic polyglutamine proteins. *NAR Mol Med*. 30;3(2).
7. Yue Y, Sablok G, He X, Hyvönen J, Dong S, Potente G, Liu Y, Szövényi P (2026). Genome assemblies of the liverwort *Blasia pusilla* uncover a well-defined pseudoautosomal region on homomorphic UV sex chromosomes. *Genome Biol*. 27:166.
8. Zanardo M, Giannattasio M, Sablok G, et al. (2023). Metabarcoding analysis of the bacterial and fungal communities during the maturation of preparation 500, used in biodynamic agriculture, suggests a rational link between horn and manure. *Environ Dev Sustain*. doi:10.1007/s10668-023-03144-w.
9. Gagnon E, Hilgenhof R, Orejuela A, McDonnell A, Sablok G, Aubriot X, Giacomini L, Gouvêa Y, Bohs L, Dodsworth S, Martine C, Poczar P, Knapp S, Särkinen T (2020). Phylogenomic data reveal hard polytomies across the backbone of the large genus *Solanum* (Solanaceae). *American Journal of Botany* 109:580-601.
10. Lehtonen S, Poczar P, Sablok G, Hyvönen J, Karger DN, Flores J (2020). Exploring the phylogeny of the marattiacean ferns. *Cladistics* 36:569-593.
11. Jike W, Li M, Zadra N, Barbaro E, Sablok G, Bertorelle G, Rota-Stabelli O, Varotto C (2020). Phylogenomic proof of recurrent demipolyploidization and evolutionary stalling of the "triploid bridge" in *Arundo* (Poaceae). *Int J Mol Sci*. 21:5247.
12. Sablok G, Amiryousefi A, He X, Hyvönen J, Poczar P (2019). Sequencing the plastid genome of giant ragweed (*Ambrosia trifida*, Asteraceae) from a herbarium specimen. *Front Plant Sci*. 10:218.
13. Jike W, Sablok G, Bertorelle G, Li M, Varotto C (2018). In silico identification and characterization of a diverse subset of conserved microRNAs in bioenergy crop *Arundo donax* L. *Sci Rep*. 8:16667.
14. Yang K, Wen X, Mudunuri S, Varma GPS, Sablok G (2019). DiffisomiRs: large-scale detection of differential isomiRs for understanding non-coding regulated stress omics in plants. *Sci Rep*. 9:1406.
15. Zanardo M, Rosselli R, Meneghesso A, Sablok G, Stevanato P, Engel M, Altissimo A, Peserico L, Dezuan V, Concheri G, Schlöter M, Squartini A (2018). Response of bacterial communities upon application of different innovative organic fertilizers in a greenhouse experiment using low-nutrient soil cultivated with *Cynodon dactylon*. *Soil Syst*. 2:52.
16. Sablok G, Hayward RJ, Davey PA, Santos RP, Schliep M, Larkum A, Pernice M, Dolferus R, Ralph PJ (2018). SeagrassDB: an open-source transcriptomics landscape for phylogenetically profiled seagrasses and aquatic plants. *Sci Rep*. 8(1):2749.
17. Mirzaei S, Mansouri M, Mohammadi-Nejad G, Sablok G (2018). Comparative assessment of chloroplast transcriptional responses highlights conserved and unique patterns across Triticeae members under salt stress. *Photosynth Res*. 136:357-369.
18. Kumar J, Sen Gupta D, Gupta S, Dubey S, Gupta P, Singh NP, Sablok G (2017). Identification, development and application of cross-species intron spanning markers in lentil. *The Crop Journal* 6:299-305.
19. Sablok G, Yang K, Rui C, Wen X (2017). tRNA derived smallRNAs: smallRNA repertoire has yet to be decoded in plants. *Front Plant Sci*. doi:10.3389/fpls.2017.01167.
20. Mattenberger F, Sabater-Muñoz B, Toft C, Sablok G, Fares MA (2017). Expression properties exhibit correlated patterns with the fate of duplicated genes, their divergence, and transcriptional plasticity in *Saccharomycotina*. *DNA Research*. doi:10.1093/dnares/dsx025.
21. Camiolo S, Sablok G, Porceddu A (2017). The evolutionary basis of translational accuracy in plants. *G3: Genes, Genomes, Genetics*. doi:10.1534/g3.117.040626.
22. Sablok G, Chen TW, Lee CC, Yang C, Gan RC, Wegrzyn JL, Porta NL, Nayak KC, Huang PJ, Varotto C, Tang P (2017). ChloroMitoCU: codon patterns across organelle genomes for functional genomics and evolutionary applications. *DNA Research*. doi:10.1093/dnares/dsw044.
23. Yang K, Sablok G, Qiao G, Nie Q, Wen X (2017). isomiR2Function: an integrated workflow for identifying microRNA variants in plants. *Front Plant Sci*. 8:322.

24. Srivastava AK, Sablok G, Hackenberg M, Deshpande U, Suprasanna P (2017). Thiourea priming enhances salt tolerance through co-ordinated regulation of microRNAs and hormones in *Brassica juncea*. *Sci Rep*. 7:4549.
25. Kumar M, Padula MP, Davey P, Pernice M, Jiang Z, Sablok G, Contreras-Porcia L, Ralph PJ (2017). Proteome analysis reveals extensive light stress-response reprogramming in the seagrass *Zostera muelleri* (Alismatales, Zosteraceae) metabolism. *Front Plant Sci*. 7:2023.
26. Ivanova Z, Sablok G, Daskalova E, Zahmanova G, Apostolova E, Yahubyan G, Baev V (2017). Chloroplast genome analysis of resurrection tertiary relict *Haberlea rhodopensis* highlights genes important for desiccation stress response. *Front Plant Sci*. 8:204.
27. Sablok G, Mudunuri SB, Edwards D, Ralph PJ (2016). Chloroplast genomics: expanding resources for an evolutionary conserved miniature molecule with enigmatic applications. *Current Plant Biology* 7-8:34-38.
28. Ahadi A, Sablok G, Hutvagner G (2017). miRTar2GO: a novel rule-based model learning method for cell line specific microRNA target prediction that integrates Ago2 CLIP-Seq and validated microRNA–target interaction data. *Nucleic Acids Research* 45:e42.
29. Sun X, Wang L, Ding J, Wang Y, Wang J, Zhang X, Che Y, Liu Z, Zhang X, Ye J, Wang J, Sablok G, Deng Z, Zhao H (2016). Integrative analysis of *Arabidopsis thaliana* transcriptomics reveals intuitive splicing mechanism for circular RNA. *FEBS Letters* 590:3510-3516.
30. Pernice M, Sinutok S, Sablok G, Commault AS, Schliep M, Macreadie PI, Rasheed MA, Ralph PJ (2016). Molecular physiology reveals ammonium uptake and related gene expression in the seagrass *Zostera muelleri*. *Marine Environmental Research* 122:126-134.
31. Zanardo M, Rosselli R, Meneghesso A, Sablok G, Stevanato P, Altissimo A, Perserico L, Dezuani V, Concheri G, Schlöter M, Squartini A (2016). Dynamics of soil prokaryotes catalyzing nitrification and denitrification in response to different fertilizers in a greenhouse experiment with *Cynodon dactylon*. *European Journal of Soil Biology* 76:83-91.
32. Sen Gupta D, Cheng P, Sablok G, Thavarajah D, Thavarajah P, Coyne C, Kumar S, Baum M, McGee R (2016). Development of a panel of unigenes derived polymorphic EST-SSR markers in lentil using public database information. *The Crop Journal* 4:425-433.
33. Davey PA, Pernice M, Sablok G, Larkum A, Lee HT, Golicz A, Edwards D, Dolferus R, Ralph P (2016). The emergence of molecular profiling and omics techniques in seagrass biology; furthering our understanding of seagrasses. *Functional Integrative Genomics* 16:465-480.
34. Lee HT, Golicz AA, Bayer PE, Jiao Y, Tang H, Paterson AH, Sablok G, Krishnaraj RR, Chan CKK, Batley J, Kendrick GA, Larkum AWD, Ralph PJ, Edwards D (2016). The genome of a southern hemisphere seagrass species (*Zostera muelleri*). *Plant Physiol*. 172:272-283.
35. Gottardini E, Cristofori A, Pellegrini E, LaPorta N, Nali C, Baldi P, Sablok G (2016). Suppression subtractive hybridization and NGS reveal differential transcriptome expression profiles in Wayfaring Tree (*Viburnum lantana* L.) treated with ozone. *Front Plant Sci*. 7:713.
36. Sablok G, Perez A, Do T, Seong TY, Sanchez C, La Porta N, Ralph PJ, Squartini A, Munoz A, Harikrishna JA (2016). PlantFuncSSR: integrating first and next generation transcriptomics for mining of SSR-functional domains markers. *Front Plant Sci*. 7:878.
37. Sablok G, Zhao H, Sun X (2016). Plant circular RNAs (circRNAs): transcriptional regulation beyond miRNAs in plants. *Molecular Plant* 9:192-194.
38. Camiolo S, Sablok G, Porceddu A (2016). Altools: a user-friendly NGS data analyser. *Biology Direct* 11:8.
39. Fu Y, Poli M, Sablok G, Wang B, Liang Y, La Porta N, Velikova V, Loreto F, Li M, Varotto C (2016). Dissection of early transcriptional responses to water stress in *Arundo donax* L. by unigene-based RNA-Seq. *Biotechnology for Biofuels* 9:54.
40. Paital B, Sablok G, Kumar S, Singh SK, Chainy GBN (2016). Investigating the conformational structure and potential site interactions of SOD inhibitors on Ec-SOD in marine mud crab *Scylla serrata*: a molecular modeling approach. *Interdiscip Sci Comput Life Sci* 8:312-318.
41. Kwasniewski M, Daszkowska-Golec A, Janiak A, Chwiałkowska K, Nowakowska U, Sablok G, Szarejko I (2016). Transcriptome analysis reveals the role of the root hairs as environmental sensors to maintain plant functions under water-deficiency conditions. *Journal of Experimental Botany* 67:1079-1094.
42. Iqbal MA, Jaiswal S, Angadi UB, Sablok G, Arora V, Kumar S, Rai A, Kumar D (2015). SBMDb: first whole genome putative microsatellite DNA marker database of sugarbeet for bioenergy and industrial applications. *Database*: bav111.
43. Srivastava S, Srivastava A, Sablok G, Deshpande T, Penna S (2015). Transcriptomics profiling of Indian mustard (*Brassica juncea*) under arsenate stress identifies key candidate genes and regulatory pathways. *Front Plant Sci*. 6:646.
44. Sablok G, Srivastava AK, Penna S, Baev V, Ralph PJ (2015). isomiRs: increasing evidences of isomiRs complexity in plant stress functional biology. *Front Plant Sci*. 6:949.
45. Min XJ, Powell B, Braessler J, Meinken J, Yu F, Sablok G (2015). Genome-wide cataloging and analysis of alternatively spliced genes in cereal crops. *BMC Genomics* 16:721.
46. Sablok G, Padma Raju GV, Mudunuri SB, Prabha R, Singh DP, Baev V, Yahubyan G, Ralph PJ, La Porta N (2015). ChloroMitoSSRDB 2.00: more genomes, more repeats, unifying SSRs search patterns and on-the-fly repeat detection. *Database*: bav084.
47. Yan X, Sablok G, Feng G, Ma J, Zhao H, Sun X (2015). nagnag: identification and quantification of NAGNAG alternative splicing using RNA-Seq data. *FEBS Letters* 15:423-428.

48. Sun X, Zuo F, Ru Y, Guo J, Yan X, Sablok G (2015). SplicingTypesAnno: annotating and quantifying alternative splicing events for RNA-Seq data. *Computer Methods and Programs in Biomedicine* 119:53-62.
49. Hu S, Sablok G, Wang B, Qu D, Barbaro E, Viola R, Li M, Varotto C (2015). Plastome organization and evolution of chloroplast genes in Cardamine species adapted to contrasting habitats. *BMC Genomics* 16:306.
50. Naydenov M, Baev V, Apostolova E, Gospodinova N, Sablok G, Gozmanova M, Yahubyan G (2015). High-temperature effect on genes engaged in DNA methylation and affected by DNA methylation in Arabidopsis. *Plant Physiology and Biochemistry* 87:102-108.
51. Nielsen DA, Pernice M, Schliep M, Sablok G, Jeffries TC, Kühl M, Wangpraseurt D, Ralph PJ, Larkum AWD (2015). Microenvironment and phylogenetic diversity of Prochloron inhabiting the surface of crustose didemnid ascidians. *Environmental Microbiology* 10:4121-4132.
52. Golicz AA, Schliep M, Lee HT, Larkum AWD, Dolferus R, Batley J, Chan CKK, Sablok G, Ralph PJ, Edwards D (2015). Genome-wide survey of the seagrass *Zostera muelleri* suggests modification of the ethylene signalling network. *Journal of Experimental Botany* 66:1489-1498.
53. La Porta N, Sablok G, Emilliani G, Hietala AM, Giovannelli A, Fontana P, Potenza E, Baldi P (2015). Identification of low temperature stress regulated transcript sequences and gene families in Italian cypress. *Molecular Biotechnology* 57:407-418.
54. Sánchez-Paz A, Muhlia-Almazan A, Saborowski R, García-Carreño F, Sablok G, Mendoza-Cano F (2014). Marine viruses: the beneficial side of a threat. *Applied Biochemistry and Biotechnology* 174:2368-2379.
55. Baev V, Milev I, Naydenov M, Vachev T, Apostolova E, Mehterov N, Gozmanva M, Minkov G, Sablok G, Yahubyan G (2014). Insight into small RNA abundance and expression in high- and low-temperature stress response using deep sequencing in Arabidopsis. *Plant Physiology and Biochemistry* 84C:105-114.
56. Sablok G, Fu Y, Bobbio V, Laura M, Rotino GL, Bagnaresi P, Allavena A, Velikova V, Viola R, Loreto F, Li M, Varotto C (2014). Fuelling genetic and metabolic exploration of C3 bioenergy crops through the first reference transcriptome of *Arundo donax* L. *Plant Biotechnology Journal* 12:554-567.
57. Baev V, Sablok G, Minkov I (2014). Next generation sequencing crowd sourcing at BIOCOMP: what promises it holds for us in future? *Journal of Computational Science* 5:325-326.
58. Jha Y, Sablok G, Fazil MHU T, Subramanian RB, Kumar S (2014). Insights into differential expression of RAB18 using in silico structural analysis and interaction with GTP binding in salinity stress condition. *Journal of Molecular Recognition* 27:521-527.
59. Stevanato P, Broccanello C, Biscarini F, Del Corvo M, Sablok G, Panella L, Stella A, Concheri G (2014). High-throughput RAD-SNP genotyping for characterization of sugar beet genotypes. *Plant Molecular Biology Reporter* 32:691-696.
60. Sablok G, Harikrishna JA, Min XJ (2013). Next generation sequencing for better understanding alternative splicing: way ahead for model and non-model plants. *Transcriptomics* 2013:e103.
61. Sablok G, Milev I, Minkov G, Minkov I, Varotto C, Yahubyan G, Baev V (2013). isomiRex: web-based identification of microRNAs, isomiR variations and differential expression using next-generation sequencing datasets. *FEBS Letters* 587:2629-2634.
62. Singh S, Sablok G, Farmer R, Singh AK, Gautam B, Kumar S (2013). Molecular dynamic simulation and inhibitor prediction of Cysteine synthase structured model as a potential drug target for Trichomoniasis. *BioMed Research International*: 390920.
63. Sablok G, Wu X, Kuo J, Nayak KC, Baev V, Varotto C, Zhou F (2012). Combinational effect of dominant mutational bias and light translational selection for translation efficiency in *Solanum lycopersicum* cv. Micro-Tom. *Genomics* 101:290-295.
64. Yaseen M, Ahmad T, Sablok G, Standardi A, Hafiz IA (2013). Role of carbon sources for in vitro plant growth & development. *Molecular Biology Reports* 40:2837-2849.
65. Kovi MR, Sablok G, Bai X, Wendell M, Rognli OA, Yu H, Xing Y (2013). Expression patterns of photoperiod and temperature regulated heading date genes in *Oryza sativa*. *Computational Biology and Chemistry* 45:36-41.
66. Tolin S, Arrigoni G, Moscatiello R, Masi A, Navazio L, Sablok G, Squartini A (2012). Naringenin-inducible proteome analysis in *Rhizobium leguminosarum* by isobaric tagging in quantitative mass spectrometry. *Proteomics* 13:1961-1972.
67. Sablok G, Mudunuri S, Patan S, Popova M, Fares M, La Porta N (2013). ChloroMitoSSRDB: open source repository of perfect and imperfect repeats in organelle genomes for evolutionary genomics. *DNA Research* 20:127-133.
68. Walters B, Lum G, Sablok G, Min XJ (2013). Genome-wide landscape of alternative splicing events in *Brachypodium distachyon*. *DNA Research* 20:163-171.
69. Ahmad T, Sablok G, Tatarinova T, Xu Q, Deng XX, Guo WW (2013). Evaluation of codon biology in Citrus and Poncirus trifoliata based on genomic features and frame corrected expressed sequence tags. *DNA Research* 20:135-150.
70. Sablok G, Sanchez-Paz A, Wu XM, Ranjan J, Kuo J, Bulla I (2012). Genome dynamics in three different geographical isolates of white spot syndrome virus (WSSV) genome. *Archives of Virology* 157:2357-2362.
71. Pérez-Quintero AL, Sablok G, Tatarinova TV, Kuo J, Conesa A, López C (2012). Computational analyses of miRNAs from transcriptome of anti-malarial plant – *Artemisia annua*. *Biotechnology Letters* 34:737-745.
72. Sablok G, Pérez-Quintero AL, Hassan M, Tatarinova TV, López C (2011). Artificial microRNAs (amiRNAs) engineering – on how microRNA-based silencing methods have affected current plant silencing research. *Biochemical and Biophysical Research Communications* 406:315-319.

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75. Sablok G, Gupta PK, Baek JM, Vazquez F, Min XJ (2011). Genome-wide survey of alternative splicing in the grass *Brachypodium distachyon* – an emerging model biosystem for plant functional genomics. *Biotechnology Letters* 33:629-636.

## Peer-Reviewed Abstracts and Posters

1. Sablok G, He X, Miranto M, Peltomaa E, Sleith R, Karol K, Delwiche C, Bell N, Paulin L, Poczar P, Hyvönen J (2019). First draft genome assembly of *Coleochaete orbicularis*. 5th Conference on Plant Genome Evolution, Spain.
2. Sablok G, He X, Miranto M, Flores JR, Peltomaa E, Sleith R, Karol K, Delwiche C, Bell N, Paulin L, Poczar P, Hyvönen J (2019). Mitogenomics of *Blasia pusilla* as a tool for agricultural productivity. 3rd Agriculture and Climate Change Conference, Budapest, Hungary, P102.
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